Exploring Wealth Disparities in Homeownership and College Education

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Problem Statement and Background:

The US has a history of systemic racism which has bled into the issues of socioeconomic still prevalent today. The effects include the wealth gap, skewed income distribution, homeownership, and a lot more in areas of economic life. Furthermore, this disproportionately affects minorities exacerbating the wealth gap across generations.

This project is aimed to analyze the wealth inequality across the US from the 1960s to 2020 with a focus on key drivers that affect the distribution such as race, homeownership, and educational attainment. Gaining an understanding of this is essential to combat the economic disparity while implementing policies and change to close the gap. In order to do this we must understand the nature or the wealth gap and its influences. By gaining insights we are able to push for initiatives that support an equitable economy in the future.

Introduction to "State_wealth_disparity" Dataset:

The dataset provides the distribution of wealth by state in the United States from the 1960s to 2020. A few of these indicators include:

- Wealth mean - mean household wealth

- Own - proportion of households that own their home
- College - proportion of households that graduated college

The data was acquired from openICPSR which is a self publishing repository for researchers to publish their research. The data was assembled by researchers from Arizona State university from their primary data– surveys conducted by the U.S. Federal Reserve.

A few ethical concerns that may arise are privacy issues. Considering the data holds sensitive information such as race, there may be bias in the data. This is due to the implications of historical racial issues which can skew the data. Despite these concerns, it is essential to recognize and address all contributing factors when manipulating the data.

Data Science Approaches

Many data science algorithms were used in conducting analyses to uncover patterns and relationships in the data. Linear regression was used to quantify the relationship between wealth inequality and its potential indicators. K-Nearest Neighbor (KNN) allowed for classification. To enhance predictive accuracy, Random Forest Classifier, a method combining multiple decisions was implemented. Additionally, K-means clustering grouped the data into meaningful clusters highlighting patterns related to wealth inequality. Furthermore, standardization was used to scale numerical features. Lastly, evaluation and statistical metrics such as mean squared error , r-squared, etc were used to assess model performance and reliability.

Results and Conclusions

Our project examines the wealth inequality across U.S. states, focusing on the relationship between wealth mean, homeownership, and college attainment. More specifically, I took a look at

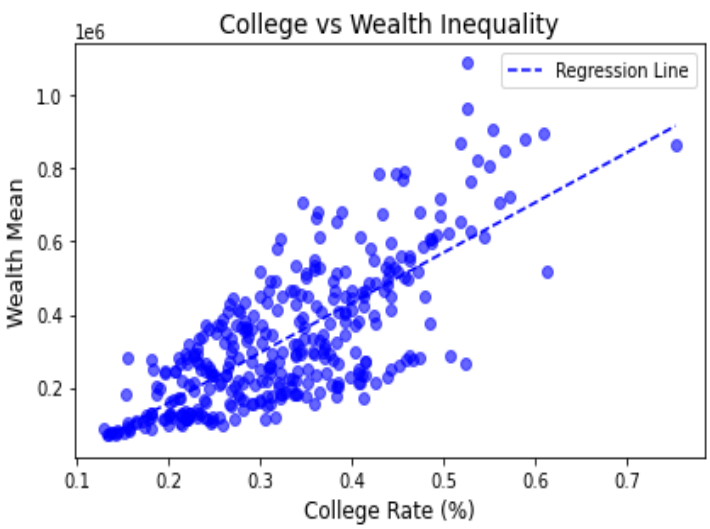
the wealth mean based on race. The findings reveal critical insights into how education, race, and homeownership correlate with wealth distribution.

Descriptive Analysis - I looked at the overall averaged of wealth, homeownership, and college

The mean wealth across states was approximately $330,234 with a standard deviation of $192,045. Households with a college education averaged 32.5% while households with homeownership averaged 63.6%. The wide standard deviation indicated variability in the wealth and indicators across states.

Regression Analysis - analyzed relationship between wealth and homeownership and college

When looking at college attainment and the wealth mean, I noticed a strong positive relationship of 0.75. The regression model is given by y=−115,016.97+1,369,786.60x. This suggests that for every 1% increase in college attainment, wealth increases by about $1,369,786.60. The r-squared of 0.54 indicates college level is a fair indicator of wealth.

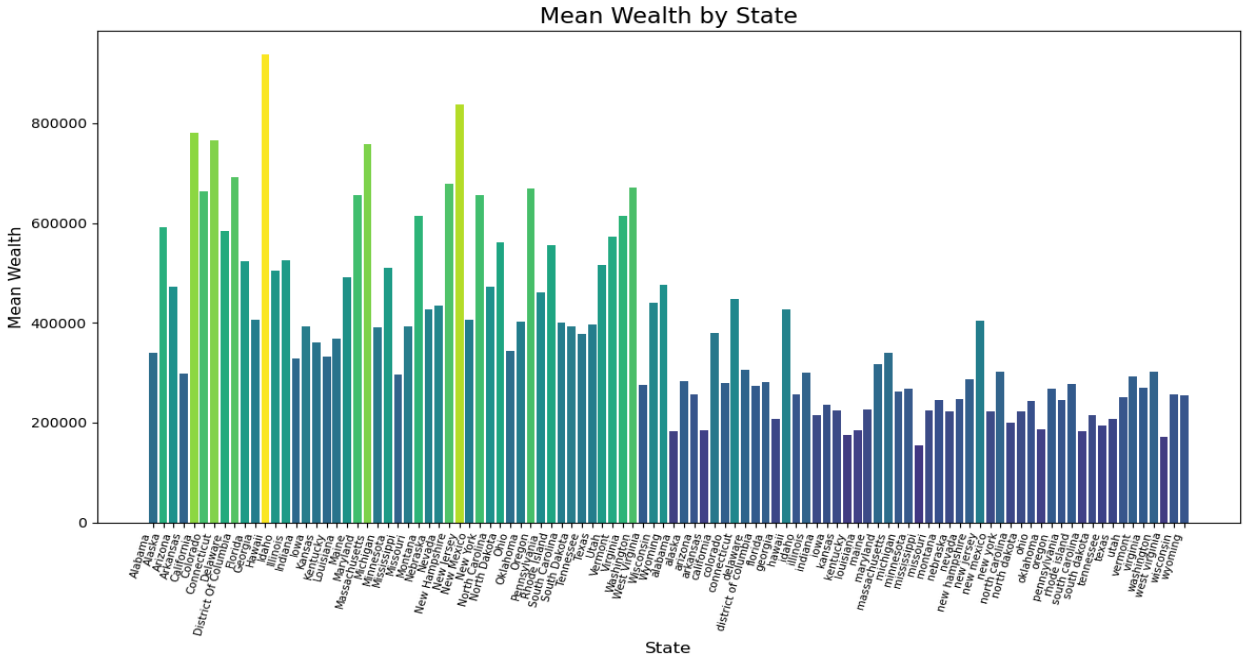


In comparison, homeownership and wealth have a much weaker connection. The correlation between the two is 0.19 suggesting a weak positive relationship. For every additional homeowner, the wealth increases by

$465,289.21 indicated by the equation y=34,387.29+465,289.21x. The r-squared of 0.04 highlights homeownership is a poor indicator of wealth.

Machine learning analysis - deeper analysis by states

After outlining the correlation/relationship between wealth mean and homeownership and college, I look into these indicators by state. Firstly, is a plot demonstrating the wealth distribution by state. As you can see below, there is variation in wealth across states. This helps identify which states contribute most to the wealth and possibly the wealth gap of the U.S.

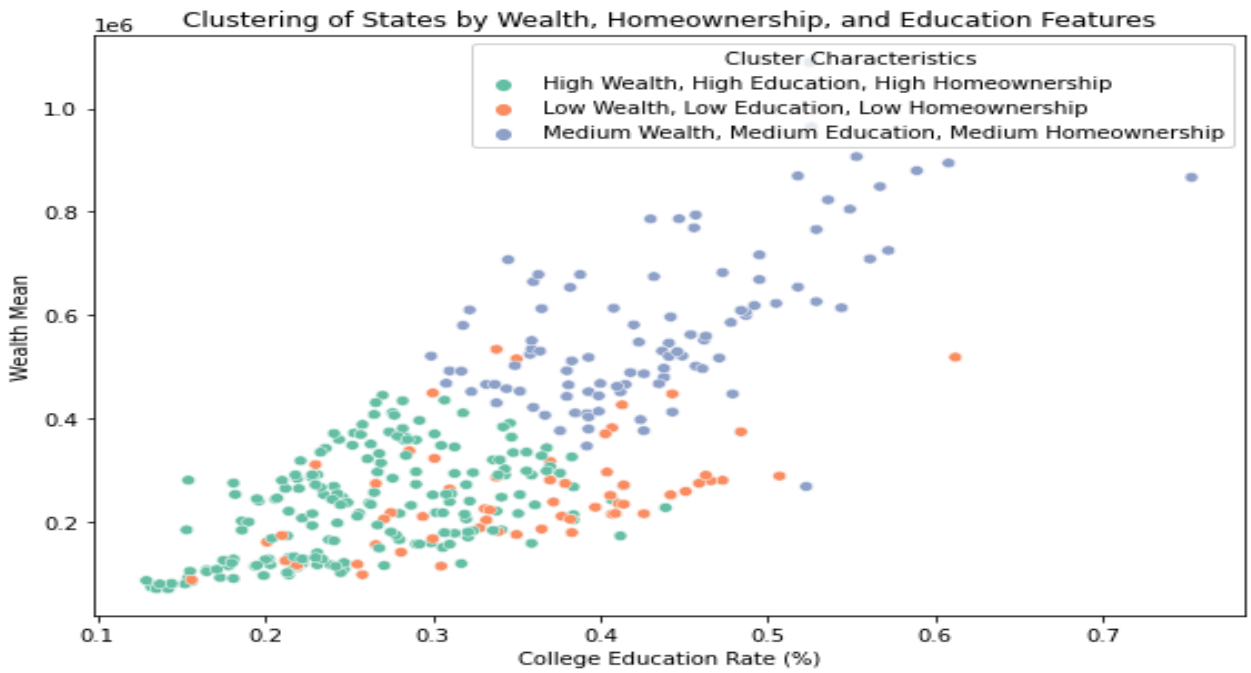


Once plotted, a KNN classifier was used to predict whether a state has high or low wealth inequality based on wealth mean, homeownership, and education rate. The model worked by classifying a state as high or low inequality based on its nearest neighbor. The model performed

an accuracy of 97%. Additional metrics include an F-1 score of 0.97 showing the model is balanced in terms of precision and recall.

After the KNN classifier, K Means clusters were utilized to group states into clusters based on features such as wealth mean, homeownership rate, and college education. The clustered allow for an even more thorough understanding of how states categorize in different levels of wealth and education. As you show below, the clusters are divided into the three ways:

- Cluster 1: states with high wealth, high education, high homeownership
- Cluster 2: states with low wealth, low education, low homeownership
- Cluster 2: states with medium wealth, medium education, medium homeownership

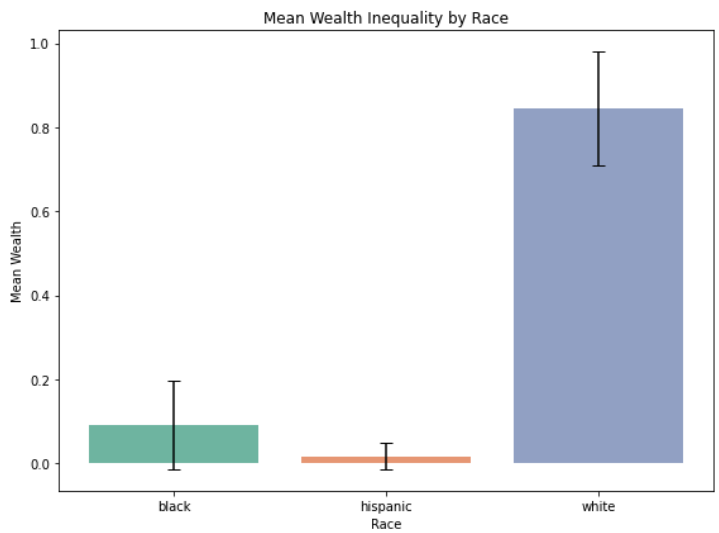


The cluster centers give average characteristics of each group should how wealth, education, and homeownership intersect in different states. It is clear that the indicators that are all high are less dispersed compared to the medium indicators.

Race-Based Analysis - included an extra factor in the analysis

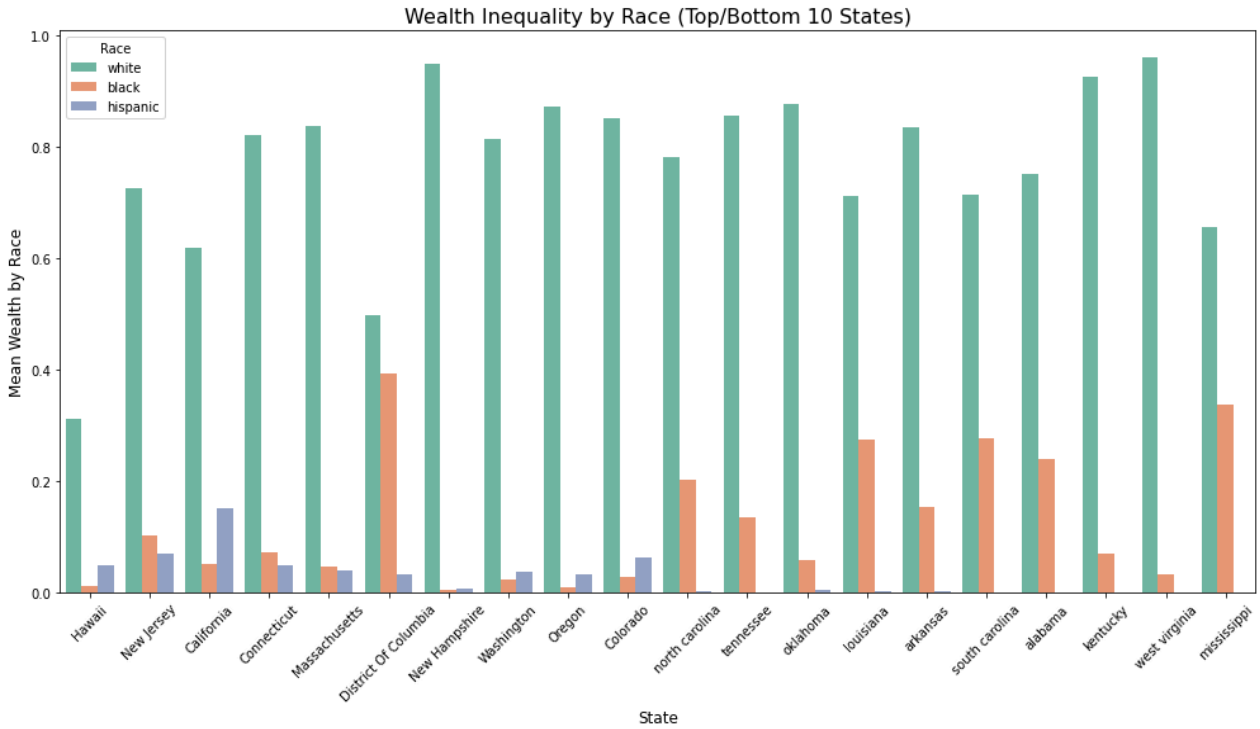
In this section of the project, I included another factor, race. I compared the relationship between race and wealth as well in states.

First, I looked at the distribution of wealth amongst races. I saw a jarring wealth gap which aligns with the racial disparities observed in the U.S. White individuals showed the highest mean wealth with an average of 84.61%. Black individuals showed a much lower health average of 9.04%.

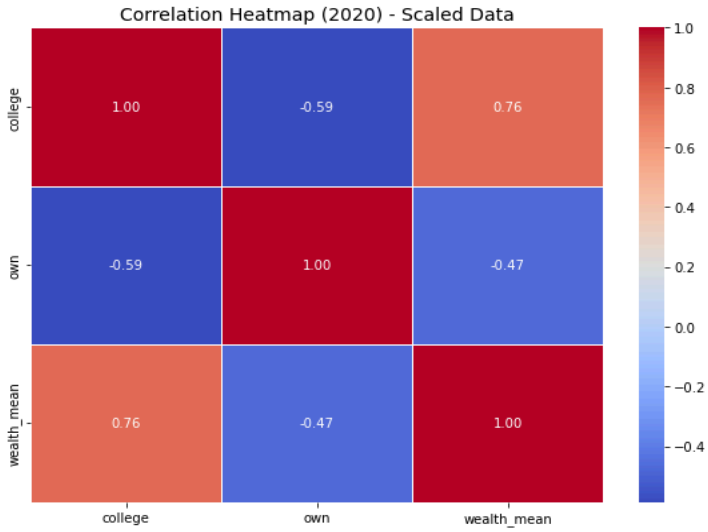


Hispanic individuals held the lowest wealth with an average of 1.73%. As shown in the bar plot, while the standard deviation for white individuals is greater, the variation in wealth among Black and Hispanic individuals includes values as low as 0%, indicating severe wealth inequality.

Further looking at the racial disparity, I look at it in relation to the 10 states with highest and lowest wealth shown below. States with the highest wealth include Hawaii, Massachusetts, New Hampshire, etc. States with the lowest wealth include Alabama, Mississippi, Arkansas, etc. Evidently, white individuals have the highest wealth as well including in states that are the least wealthiest of the states. Black individuals follow with significantly lower wealth averages, alternating with Hispanic individuals in certain states, such as Washington and Oregon.



2020 wealth analysis - looked at all previous analysis current day



In order to understand the wealth distribution, I decided to do a comparison to the most recent year in the data set, 2020, focusing on how wealth maps against college attainment and homeownership. This was done via a heatmap describing the correlation between wealth and its indicators. Wealth has a strong positive correlation with college of 76% and a negative moderate correlation of -47% with homeownership. This suggests college education plays a significant role in wealth today.

Conclusion - final summary of analysis

In conclusion, the analysis reveals a several key insights:

- Wealth disparity and college education are closely associated, and states with greater rates of college education also typically have higher wealth mean values
- With only a slight positive association to wealth mean, the homeownership rate has a smaller relationship with wealth disparity
- Wealth disparity is significantly influenced by race, with white people typically having greater wealth than black and Hispanic people
- The 2020 analysis also highlights how crucial education is to the distribution of wealth in current day
- To better understand how various states group based on wealth-related characteristics, the KNN classification model and KMeans clustering offer practical methods for classifying and forecasting patterns of wealth disparity

Future Work

All things considered, this analysis highlights the enduring wealth gap and the ways homeownership, education, and race shape economic outcomes. This research deepens our understanding of wealth inequality and points to the need for targeted policies to promote equitable access to education, housing, and financial services. Future work could explore employment opportunities and generational wealth to uncover solutions that ensure fair economic opportunities for all. As we progress economically, I hope to see initiatives implemented that target an equitable socio economic future.

Sources cited

Suss, Joel, Connor, Dylan, and Kemeny, Tom. GEOWEALTH-US: Spatial wealth inequality data for the United States, 1960-2020. Ann Arbor, MI: Inter-university Consortium for Political and Social Research [distributor], 2024-02-22. https://doi.org/10.3886/E192306V4